

7

Two more general theorems

The theorems proved in this chapter are separated from those of Chapter 6 on account of their slightly more technical nature and the fact that we do not use them in an essential way in what follows. Both theorems assert the exactness of certain sequences. The first attempts to answer the question: given the homology groups of two subcomplexes L_1, L_2 of a simplicial complex K (not necessarily disjoint), what are the homology groups of $L_1 \cup L_2$? They obviously depend on the homology groups of $L_1 \cap L_2$ as well, and moreover on the way in which L_1 and L_2 are stuck together – for example two cylinders can be stuck together to give either a torus or a Klein bottle (see 7.5). The result is not an explicit formula for $H_p(L_1 \cup L_2)$ but an exact sequence which, with luck, will give a good deal of information. An example where it does not give quite enough information to determine a homology group of $L_1 \cup L_2$ is mentioned in 7.6(3).

The other theorem proved in this chapter, the exactness of the ‘homology sequence of a triple’, is a generalization of the homology sequence of a pair (see 6.1).

The Mayer–Vietoris sequence

Let K be an oriented simplicial complex of dimension n , and let L_1, L_2 be subcomplexes of K with $L_1 \cup L_2 = K$. Write $L_1 \cap L_2 = L$ and, as usual, regard $C_p(L)$, $C_p(L_1)$ and $C_p(L_2)$ as subgroups of $C_p(K)$, for each p .

Definitions 7.1. (1) Define a homomorphism

$$\phi : H_p(L) \rightarrow H_p(L_1) \oplus H_p(L_2)$$

by

$$\{z\}_L \rightarrow (\{z\}_{L_1}, -\{z\}_{L_2}).$$

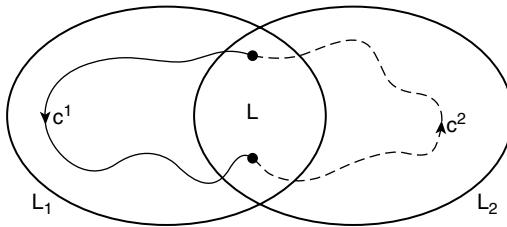
(2) Define a homomorphism

$$\psi : H_p(L_1) \oplus H_p(L_2) \rightarrow H_p(K)$$

by

$$(\{z^1\}_{L_1}, \{z^2\}_{L_2}) \rightarrow \{z^1 + z^2\}_K.$$

(3) Suppose that $z \in Z_p(K)$, and that $z = c^1 + c^2$ where $c^1 \in C_p(L_1)$ and $c^2 \in C_p(L_2)$. Then $0 = \partial z = \partial c^1 + \partial c^2$, so that $\partial c^1 = -\partial c^2$ and both ∂c^1 and ∂c^2 must belong to $C_{p-1}(L)$ – indeed to $Z_{p-1}(L)$ since $\partial \circ \partial = 0$.



Define a homomorphism

$$\Delta : H_p(K) \rightarrow H_{p-1}(L)$$

by

$$\{z\}_K \rightarrow \{\partial c^1\}_L.$$

To prove this makes sense, suppose that $\{w\}_K = \{z\}_K$, and that $w = c^3 + c^4$ where $c^3 \in C_p(L_1)$ and $c^4 \in C_p(L_2)$. We have to show $\{\partial c^1\}_L = \{\partial c^3\}_L$. Now $z - w = \partial c$ for some $c \in C_{p+1}(K)$, so that writing $c = c^5 + c^6$ ($c^5 \in C_{p+1}(L_1)$, $c^6 \in C_{p+1}(L_2)$), we have $c^1 - c^3 - \partial c^5 = -(c^2 - c^4 - \partial c^6)$. Accordingly both sides of this equation belong to $C_p(L)$, whence

$$\partial(c^1 - c^3 - \partial c^5) = \partial c^1 - \partial c^3 \in B_{p-1}(L),$$

i.e. $\{\partial c^1\}_L = \{\partial c^3\}_L$ as required.

The verifications that (1) and (2) also make sense are left to the reader. We now fit the three homomorphisms ϕ , ψ , Δ into an exact sequence, called the *Mayer–Vietoris sequence* of the triad $(K; L_1, L_2)$.

Theorem 7.2. *The following sequence is exact.*

$$\begin{aligned} 0 \rightarrow H_n(L) &\xrightarrow{\phi} H_n(L_1) \oplus H_n(L_2) \xrightarrow{\psi} H_n(K) \xrightarrow{\Delta} H_{n-1}(L) \\ &\xrightarrow{\phi} H_{n-1}(L_1) \oplus H_{n-1}(L_2) \xrightarrow{\psi} \dots \xrightarrow{\Delta} H_0(L) \xrightarrow{\phi} H_0(L_1) \oplus H_0(L_2) \\ &\xrightarrow{\psi} H_0(K) \rightarrow 0. \end{aligned}$$

Proof (1) Exactness at $H_p(L)$. Let $z \in Z_{p+1}(K)$ and $z = c^1 + c^2$ where $c^1 \in C_{p+1}(L_1)$, $c^2 \in C_{p+1}(L_2)$. Then

$$\phi \Delta \{z\} = (\{\partial c^1\}_{L_1}, -\{\partial c^1\}_{L_2}) = (\{\partial c^1\}_{L_1}, \{\partial c^2\}_{L_2}) = (0, 0).$$

Hence $\text{Im } \Delta \subset \text{Ker } \phi$.

Conversely, suppose that $u \in Z_p(L)$ satisfies $\phi\{u\} = (0, 0)$, i.e. $\{u\}_{L_1} = 0$ and $\{u\}_{L_2} = 0$. Thus $u = \partial x^1 = \partial x^2$, say, so $\partial(x^1 - x^2) = 0$, i.e. $x^1 - x^2 \in Z_p(K)$. Furthermore $\Delta\{x^1 - x^2\}_K = \{\partial x^1\}_L = \{u\}_L$. Hence $\text{Ker } \phi \subset \text{Im } \Delta$.

(2) Exactness at $H_p(L_1) \oplus H_p(L_2)$. Let $z \in Z_p(L)$; then $\psi\phi\{z\} = \{z - z\}_L = 0$. Hence $\text{Im } \phi \subset \text{Ker } \psi$.

Conversely, suppose that $(\{u^1\}, \{u^2\}) \in H_p(L_1) \oplus H_p(L_2)$ lies in $\text{Ker } \psi$, i.e. that $\{u^1 + u^2\}_K = 0$. Thus $u^1 + u^2 = \partial x$ for some $x \in C_{p+1}(K)$. Let $\rho_\alpha : C_{p+1}(K) \rightarrow C_{p+1}(L_\alpha)$, $\alpha = 1, 2$ be the restriction homomorphisms which change to zero the coefficient of any $(p+1)$ -simplex not in L_α ; I claim that $u^1 - \partial\rho_1 x \in Z_p(L)$ and that $\phi\{u^1 - \partial\rho_1 x\} = (\{u^1\}, \{u^2\})$. To see this, note that

$$x - \rho_1 x - \rho_2 x \in C_{p+1}(L)$$

so that

$$\partial x - \partial\rho_1 x - \partial\rho_2 x \in B_p(L)$$

i.e.

$$(u^1 - \partial\rho_1 x) + (u^2 - \partial\rho_2 x) \in B_p(L).$$

But the first bracket belongs to $C_p(L_1)$ and the second to $C_p(L_2)$; hence both belong to $C_p(L)$ and indeed since u^1 and u^2 are cycles both brackets belong to $Z_p(L)$. Finally $u^\alpha - \partial\rho_\alpha x \sim u^\alpha$ on L_α ($\alpha = 1, 2$) and $-(u^1 - \partial\rho_1 x) \sim u^2 - \partial\rho_2 x$ on L , and hence on L_2 . Applying the definition of ϕ the required result follows. Thus $\text{Ker } \psi \subset \text{Im } \phi$. (The reader may like to draw a diagram to illustrate this inclusion, say in the case $p = 1$.)

(3) Exactness at $H_p(K)$. Let $(\{z^1\}, \{z^2\}) \in H_p(L_1) \oplus H_p(L_2)$. Then $\Delta\psi(\{z^1\}, \{z^2\}) = \{\partial z^1\}_L = 0$ since z^1 is a cycle. Hence $\text{Im } \psi \subset \text{Ker } \Delta$.

Conversely, suppose that $u \in Z_p(K)$ satisfies $\Delta\{u\} = 0$, i.e. writing $u = c^1 + c^2$ in the usual way, that $\{\partial c^1\}_L = 0$. This means that $\partial c^1 = \partial x$ for some $x \in C_p(L)$. Then $c^1 - x \in Z_p(L_1)$ and $c^2 + x \in Z_p(L_2)$ since u is a cycle. Since

$$\psi(\{c^1 - x\}_{L_1}, \{c^2 + x\}_{L_2}) = \{c^1 + c^2\}_K = \{u\}_K$$

this shows that $\text{Ker } \Delta \subset \text{Im } \psi$. □

Remark 7.3. There is a reduced form of the Mayer–Vietoris sequence in which zero-dimensional homology groups are replaced by reduced groups (4.8) and Δ , ϕ and ψ are suitably altered at the lower end of the sequence. The resulting sequence is still exact.

Examples 7.4. (1) If L is empty, then K is the union of disjoint subcomplexes L_1, L_2 and from the Mayer–Vietoris sequence

$$H_p(K) \cong H_p(L_1) \oplus H_p(L_2)$$

for all p . This result extends in an obvious way to the union of any number of disjoint subcomplexes, and it is not hard to give a direct proof.

(2X) Suppose that L is connected. Then

$$\phi : H_0(L) \rightarrow H_0(L_1) \oplus H_0(L_2)$$

is monic (injective). (If L is empty there is nothing to prove; otherwise let v be a vertex of L so that $\{v\}$ generates the infinite cyclic group $H_0(L)$. Now

$$\phi\{v\} = (\{v\}_{L_1}, -\{v\}_{L_2}) \neq (0, 0)$$

and since $H_0(L_1) \oplus H_0(L_2)$ has no non-zero element of finite order this is the result.)

(3X) Suppose that L consists of a single point. Then $H_p(L) = 0$ for $p > 0$ so that from the Mayer–Vietoris sequence $H_p(K) \cong H_p(L_1) \oplus H_p(L_2)$ for $p > 1$. From (2) it follows that this holds for $p = 1$ as well. What about $p = 0$?

(4X) Suppose that L consists of two points, v and w say. Show that

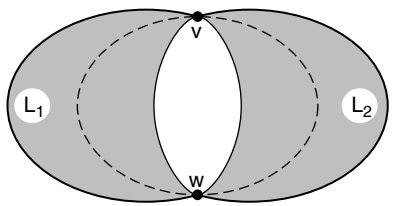
$$\phi : H_0(L) \rightarrow H_0(L_1) \oplus H_0(L_2)$$

has infinite cyclic kernel generated by $\{v - w\}$ if v and w lie in the same component of L_1 and in the same component of L_2 , and is monic (injective)

otherwise. (Note that if v and w lie in different components of L_1 then $\lambda v + \mu w \sim 0$ on L_1 implies $\lambda = \mu = 0$. Compare the proof of 4.11.) Hence show that

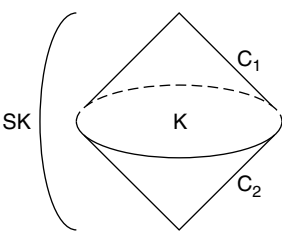
$$\begin{aligned} H_p(K) &\cong H_p(L_1) \oplus H_p(L_2), \quad p > 1 \\ H_1(K) &\cong H_1(L_1) \oplus H_1(L_2) \oplus A \end{aligned}$$

where $A = 0$ or $A \cong \mathbb{Z}$. In particular $A \cong \mathbb{Z}$ when K is connected, and the diagram shows where the extra 1-cycle comes from. Find examples to illustrate the case $A = 0$.



(5X) Some earlier results can be reproved using the Mayer–Vietoris sequence, for instance 6.11(4) (homology groups of S^n), 6.6 (collapsing) and 6.12 (effect of removing a 2-simplex from a closed surface). The reader is invited to try these examples, and to look for others in Chapter 6 amenable to this alternative treatment.

(6X) Let K be an oriented simplicial complex in $\mathbb{R}^N$, entirely contained in the subspace $x_1 = 0$ (where $(x_1, \dots, x_N)$ are coordinates in $\mathbb{R}^N$). Let C_1 be the cone on K with vertex $(1, 0, \dots, 0)$ and C_2 the cone on K with vertex $(-1, 0, \dots, 0)$, so that $C_1 \cap C_2 = K$. We call $C_1 \cup C_2$ a *suspension* of K and denote it by SK . Show that

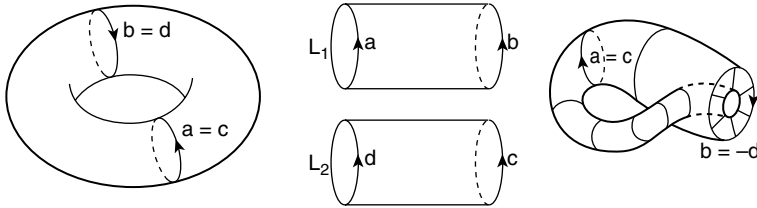


$$\begin{aligned} H_p(SK) &\cong H_{p-1}(K) \quad (p > 1) \\ H_1(SK) &\cong \widetilde{H}_0(K). \end{aligned}$$

Which closed surfaces, if any, are suspensions of graphs?

The remaining examples will be of a more informal nature than those given above, in that we shall not worry very much about the triangulations of the objects in question but rather witness the Mayer–Vietoris sequence in action by sticking together simple objects to make more complicated ones. In this way I hope that the geometrical content of the sequence will become clearer.

Example 7.5. If we identify the ends of two cylinders, the result can be either a torus or a Klein bottle.



We shall assume the following: any triangulation L_1 of a cylinder has $H_1(L_1) \cong \mathbb{Z}$, with generator a cycle going round one end; $H_2(L_1) = 0$. (Compare 4.17(1); the result could indeed be proved by a collapsing argument.) Now let us see whether the Mayer–Vietoris sequence enables us to calculate the homology groups of a torus and a Klein bottle, which are already known from 6.13. Note that in either case $L_1 \cap L_2 = L$ consists of two disjoint circles.

$$\begin{array}{ccccccc}
 H_2(L_1) \oplus H_2(L_2) & \xrightarrow{\psi_2} & H_2(K) & \xrightarrow{\Delta_2} & H_1(L) & \xrightarrow{\phi_1} & H_1(L_1) \oplus H_1(L_2) \xrightarrow{\psi_1} \\
 0 & & 0 & & \mathbb{Z} \oplus \mathbb{Z} & & \mathbb{Z} \oplus \mathbb{Z} \\
 \xrightarrow{\psi_1} & H_1(K) & \xrightarrow{\Delta_1} & H_0(L) & \xrightarrow{\phi_0} & H_0(L_1) \oplus H_0(L_2) & \xrightarrow{\psi_0} H_0(K) \rightarrow 0. \\
 & & & \mathbb{Z} \oplus \mathbb{Z} & & \mathbb{Z} \oplus \mathbb{Z} &
 \end{array}$$

Consider

$$\phi_1 : H_1(L) \rightarrow H_1(L_1) \oplus H_1(L_2).$$

The groups are

$$\mathbb{Z} \oplus \mathbb{Z} \qquad \mathbb{Z} \oplus \mathbb{Z}$$

with generators

$$\begin{array}{lcl}
 \{a\}, \{b\}_L & (\{a\}_{L_1}, 0), (0, \{c\}_{L_2}) \\
 = & \alpha_1, \quad \alpha_2 & \text{say.}
 \end{array}$$

Now

$$\begin{aligned}\phi_1\{a\}_L &= (\{a\}_{L_1}, -\{a\}_{L_2}) = (\{a\}_{L_1}, -\{c\}_{L_2}) = \alpha_1 - \alpha_2 \\ \phi_1\{b\}_L &= (\{b\}_{L_1}, -\{b\}_{L_2}) = (\{a\}_{L_1}, \pm\{c\}_{L_2}) = \alpha_1 \pm \alpha_2\end{aligned}$$

where the sign is + or − for the Klein bottle or torus respectively.
From this it follows that

$$H_2(K) \cong \text{Ker } \phi_1 \cong \begin{cases} \mathbb{Z} & \text{generator } \{a - b\}_L \text{ for torus} \\ 0 & \text{for Klein bottle.} \end{cases}$$

A generator for $H_2(K)$ in the case of the torus is any element going to $\{a - b\}_L$, under Δ_2 . Such an element is the homology class of the sum of all 2-simplexes of K , coherently oriented.

A similar examination of ϕ_0 reveals that $\text{Im } \Delta_1 = \text{Ker } \phi_0 \cong \mathbb{Z}$, with generator $\{v^1 - v^2\}_L$ where v^1, v^2 are vertices in the two components of L .

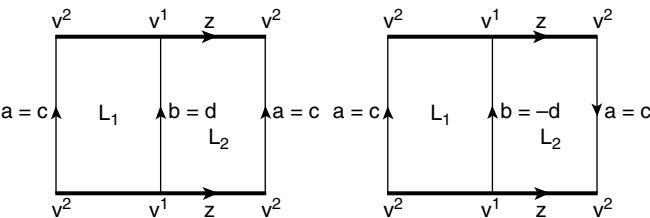
Now there is a short exact sequence

$$0 \rightarrow \text{Im } \psi_1 \xrightarrow{\text{inclusion}} H_1(K) \xrightarrow{\Delta_1} \text{Im } \Delta_1 \rightarrow 0.$$

Since $\text{Im } \Delta_1 \cong \mathbb{Z}$, this sequence splits (A.19). Also
 $\text{Im } \psi_1 \cong (H_1(L_1) \oplus H_1(L_2))/\text{Im } \phi_1$ by the homomorphism theorem (A.15).
This latter group has presentation

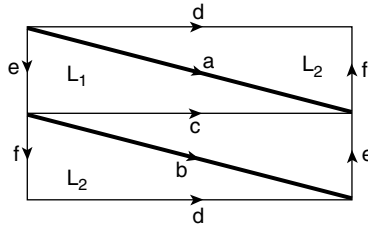
$$\begin{pmatrix} \alpha_1, \alpha_2 \\ \alpha_1 - \alpha_2 \\ \alpha_1 \pm \alpha_2 \text{ (+ for Klein bottle, - for torus)} \end{pmatrix}$$

and so is isomorphic to $\mathbb{Z}_2$ (generator $\alpha_1 + \text{Im } \phi_1$) or $\mathbb{Z}$ (generator $\alpha_1 + \text{Im } \phi_1$) according as the sign is + or −. A generator for $\text{Im } \psi_1$ is therefore $\psi_1(\alpha_1)$ in either case, i.e. $\{a\}_K$. An element of $H_1(K)$ going under Δ_1 to $\{v^1 - v^2\}_L$ is $\{z\}$, where z is indicated heavily on the diagrams below.



Thus $H_1(\text{torus}) \cong \mathbb{Z} \oplus \mathbb{Z}$; $H_1(\text{Klein bottle}) \cong \mathbb{Z} \oplus \mathbb{Z}_2$ with generators $\{z\}$ and $\{a\}$ in each case.

Examples 7.6. (1X) In the spirit of 7.5, split a Klein bottle into two Möbius bands and calculate the homology groups of the Klein bottle from those of the Möbius bands. L is a single circle, drawn heavily.



Assume that $H_1(L_1) \cong \mathbb{Z}$ (generator $\{c\}$); $H_1(L_2) \cong \mathbb{Z}$ (generator $\{d\}$); $H_1(L) \cong \mathbb{Z}$ (generator $\{a + b\}$); $a + b \sim 2c$ on L_1 ; $a + b \sim 2d$ on L_2 .

(2X) Let M_1 and M_2 be two orientable closed surfaces and let L_1, L_2 be obtained by removing a 2-simplex from, respectively, M_1 and M_2 . Then $M = L_1 \cup L_2$ is a connected sum $M_1 \# M_2$ (2.9). Use the Mayer–Vietoris sequence to show that

$$H_1(M) \cong H_1(M_1) \oplus H_1(M_2).$$

[Hints. Show that $H_1(L_1) \cong H_1(M_1)$, either by quoting 6.12 or by using the Mayer–Vietoris sequence. Then use the sequence again to show $H_1(M) \cong H_1(L_1) \oplus H_1(L_2)$.] Thus once the first homology group of any torus is known this reproduces 6.13(2), at least up to isomorphism.

(3X) The reason why (2) works is that the boundary of the ‘missing’ 2-simplex is homologous to zero on L_1 , and on L_2 . If we stick together a torus and a projective plane, say, this is no longer the case. Make the following *assumption*: $L_2 =$ triangulation of a projective plane, minus a 2-simplex, has $H_1(L_2) \cong \mathbb{Z}$ and the boundary of the ‘missing’ 2-simplex represents twice a generator. Now use the Mayer–Vietoris sequence to calculate the first homology group of the connected sum in question. The above assumption follows, of course, from the collapsing arguments of Chapter 6 – but then so does the final result! Removing a 2-simplex from a triangulation of a projective plane does in fact give a triangulation of a Möbius band, so the assumption is related to a calculation in 4.15.

It is worth noting that the Mayer–Vietoris sequence cannot by itself be used to prove the assumption, nor even that $H_1(L_2) \cong \mathbb{Z}$ given $H_1(P) = \mathbb{Z}_2$ and $H_2(P) = 0$. For the sequence only provides us with a short exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow H_1(L_2) \rightarrow \mathbb{Z}_2 \rightarrow 0$$

from which $H_1(L_2)$ could be isomorphic either to $\mathbb{Z}$ or to $\mathbb{Z} \oplus \mathbb{Z}_2$. Compare A.20(2).

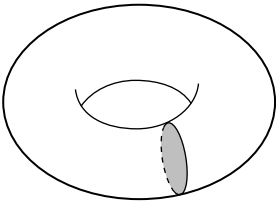
By writing a general non-orientable closed surface kP as

$$\begin{aligned} P \# \tfrac{1}{2}(k-1)T & \quad k \text{ odd} \\ \text{or } 2P \# \tfrac{1}{2}(k-2)T & \quad k \text{ even} \end{aligned}$$

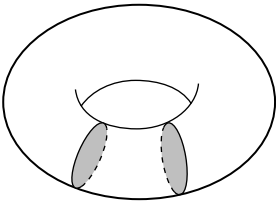
(compare the discussion following 2.10) the above result can be used to calculate $H_1(kP)$, making use of the above assumption and an analogous one for $2P =$ Klein bottle. Can the Mayer–Vietoris sequence be used to calculate $H_1(2P)$ using just the above assumption about P ?

(4X) Again in the spirit of 7.5, calculate the homology groups of the following objects.

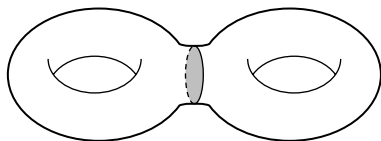
Union of a torus and a disk spanning a meridian circle. (The disk has the homology groups of a point, since (3.32) it collapses to a point.)



Union of a torus and two disks spanning parallel meridian circles.



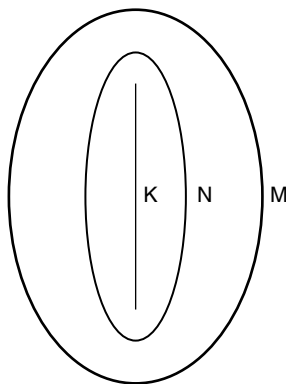
Union of a double torus and a disk spanning the ‘waist’.



★ Homology sequence of a triple

The theorem about to be proved generalizes 6.1, the exactness of the homology sequence of a pair. It will be used in Chapter 9, but not in an essential way, and is included here for completeness. There are several ways of proving the theorem: undoubtedly the ‘right’ way is to use some purely algebraic machinery such as that in 4.36 et seq. to deduce the result from an exact sequence of chain complexes; or an ad hoc proof analogous to that of 6.1 can be constructed. We shall adopt a third alternative, and deduce it from 6.1 itself, together with one of the more picturesque results in the theory of exact sequences, known as the Braid Theorem (see 7.9 below).

The notation here is chosen to be close to that in Chapter 9, where the theorem will be used. Let K and N be subcomplexes of a simplicial complex M , and suppose that $K \subset N$.



Thus $K \subset N \subset M$. There are three pairs here: (N, K) , (M, K) and (M, N) . The object is to fit their relative homology groups into an exact sequence. As usual, all chain groups are regarded as subgroups of the chain groups of M .

Definitions 7.7. Define $\alpha = \alpha_p : H_p(N, K) \rightarrow H_p(M, K)$ by $\alpha\{z\}_{N,K} = \{z\}_{M,K}$ for $z \in Z_p(N, K)$.

Define

$$\beta = \beta_p : H_p(M, K) \rightarrow H_p(M, N)$$

by

$$\beta\{z\}_{M,K} = \{jz\}_{M,N} \text{ for } z \in Z_p(M, K)$$

where

$$j : C_p(M) \rightarrow C_p(M, N)$$

is the usual homomorphism.

Define

$$\gamma = \gamma_p : H_p(M, N) \rightarrow H_{p-1}(N, K)$$

by

$$\gamma\{z\}_{M,N} = \{j'\partial z\}_{N,K} \quad z \in Z_p(M, N)$$

where

$$j' : C_{p-1}(N) \rightarrow C_{p-1}(N, K)$$

is the usual homomorphism.

The verifications that these are well-defined are left to the reader.

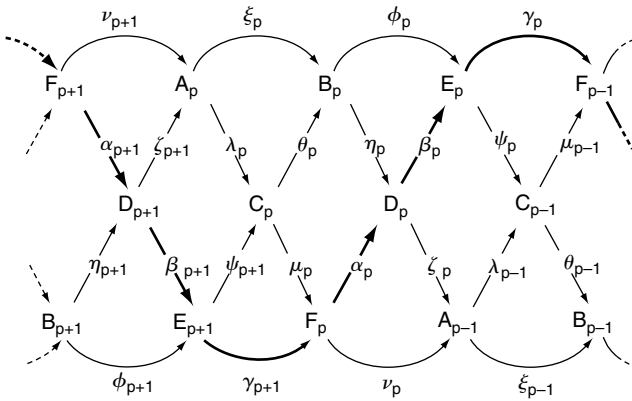
Theorem 7.8. *If M has dimension n , then the sequence*

$$\begin{aligned} 0 \rightarrow H_n(N, K) \xrightarrow{\alpha} H_n(M, K) \xrightarrow{\beta} H_n(M, N) \xrightarrow{\gamma} H_{n-1}(N, K) \xrightarrow{\alpha} \cdots \\ \cdots \rightarrow H_1(M, N) \xrightarrow{\gamma} H_0(N, K) \xrightarrow{\alpha} H_0(M, K) \xrightarrow{\beta} H_0(M, N) \rightarrow 0 \end{aligned}$$

is exact. It is called the homology sequence of the triple (M, N, K) .

Note that if $K = \emptyset$ this reduces to the homology sequence of the pair (M, N) . As mentioned above the proof will depend on 6.1 and a purely algebraic result on exact sequences, which is as follows.

Consider the diagram of abelian groups and homomorphisms below.



There are four ‘sine-wave’ sequences in the diagram, in which the homomorphisms are, respectively, labelled $\lambda, \mu, \nu; \xi, \eta, \zeta; \theta, \phi, \psi$; and α, β, γ . Apart from α, β , and γ , the notation in this theorem has no connexion with previous usage – sheer pressure of notation has necessitated the re-use of several symbols. The ‘regions’ of the diagram are three- or four-sided, and to say that the diagram ‘commutes’ is to say that two alternative routes from one corner group of a region to another are always the same homomorphism – thus $\eta_p \circ \theta_p = \alpha_p \circ \mu_p$ for all p , etc.

7.9. Braid theorem Suppose that the above diagram commutes and that the first three sequences mentioned above are exact. Suppose also that $\beta_p \circ \alpha_p = 0$ for all p . Then the sequence of α ’s, β ’s and γ ’s is exact.

Proof The proof is of a kind called *diagram-chasing* and we shall only give about half of it, leaving the reader to pursue the rest. It is enough to prove exactness at F_p, D_p and E_p .

(1) Exactness at F_p

$$\alpha_p \circ \gamma_{p+1} = \alpha_p \circ \mu_p \circ \psi_{p+1} = \eta_p \circ \theta_p \circ \psi_{p+1} = 0$$

by exactness of the θ, ϕ, ψ sequence.

Hence $\text{Im } \gamma_{p+1} \subset \text{Ker } \alpha_p$.

Conversely suppose $\alpha_p f = 0, f \in F_p$. Then $0 = \zeta_p \alpha_p f = \nu_p f$; hence $f = \mu_p c$ for some $c \in C_p$. Thus $0 = \alpha_p \mu_p c = \eta_p \theta_p c$, so $\theta_p c \in \text{Ker } \eta_p = \text{Im } \xi_p$. Hence $\theta_p c = \xi_p a$ for some $a \in A_p$. We now have $\theta_p(\lambda_p a) = \xi_p a = \theta_p c$; hence

$c - \lambda_p a \in \text{Ker } \theta_p = \text{Im } \psi_{p+1}$. This gives $c - \lambda_p a = \psi_{p+1}e$ for some $e \in E_{p+1}$, and

$$\gamma_{p+1}e = \mu_p \psi_{p+1}e = \mu_p(c - \lambda_p a) = \mu_p c = f.$$

Hence $\text{Ker } \alpha_p \subset \text{Im } \gamma_{p+1}$.

(2) Exactness at D_p . One half of this is given; for the other half suppose $\beta_p d = 0$ for some $d \in D_p$. Then $0 = \psi_p \beta_p d = \lambda_{p-1} \zeta_p d$. Hence $\zeta_p d \in \text{Ker } \lambda_{p-1} = \text{Im } \nu_p$, so $\zeta_p d = \nu_p f$ for some $f \in F_p$. Thus $\zeta_p \alpha_p f = \nu_p f = \zeta_p d$, so $d - \alpha_p f \in \text{Ker } \zeta_p = \text{Im } \eta_p$. This gives $d - \alpha_p f = \eta_p b$ for some $b \in B_p$, and since $\phi_p b = \beta_p \eta_p b = 0$ (since $\beta_p d = 0$), $b \in \text{Ker } \phi_p = \text{Im } \theta_p$, and $b = \theta_p c$ for some $c \in C_p$. Thus

$$\alpha_p(\mu_p c + f) = \eta_p \theta_p c + \alpha_p f = \eta_p b + d - \eta_p b = d.$$

Hence $\text{Ker } \beta_p \subset \text{Im } \alpha_p$.

The proof of exactness at E_p is left to the reader. $\square$

To deduce that the homology sequence of a triple is exact we make the following substitutions:

$$\begin{aligned} A_p &= H_p(K), & B_p &= H_p(M), & C_p &= H_p(N), & D_p &= H_p(M, K), \\ E_p &= H_p(M, N), & F_p &= H_p(N, K). \end{aligned}$$

The homomorphisms being the usual ones, and $\alpha_p, \beta_p, \gamma_p$ being as in 7.7, the three sequences named first in the preamble to 7.9 become the homology sequences of the pairs (N, K) , (M, K) and (M, N) respectively, while the fourth sequence becomes the homology sequence of the triple (M, N, K) . The details of checking that the hypotheses of the Braid Theorem are satisfied are left to the reader. $\square$

Example 7.10. (X) In the notation of 7.8, suppose that $N \searrow K$. Using $H_p(N, K) = 0$ (see the proof of 6.6) and 7.8 deduce that $H_p(M, K) \cong H_p(M, N)$ for all p . What are the corresponding results (i) if $M \searrow N$; (ii) if $M \searrow K$? ★